

Acoustic FMCW Radar — Vital Sign Tracking Simulation

Comprehensive DSP Analysis, Range Profiling, Stress Tests & Benchmark Report

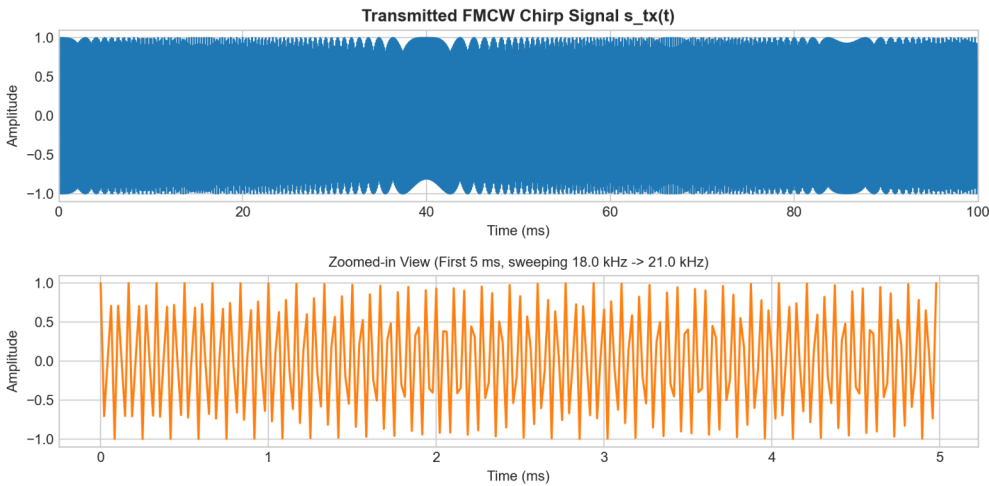
SYSTEM & RADAR PARAMETERS:

• Sampling Frequency: 48000 Hz		• Chirp Bandwidth: 3.0 kHz (18.0 - 21.0 kHz)		
• Chirp Duration Tc: 100 ms		• Frame Interval Tframe: 150 ms		• Chirps: 60 (9.0 s duration)
• Speed of Sound: 343.0 m/s		• Carrier Center Frequency fc: 19.50 kHz		• Wavelength lambda: 17.59 mm

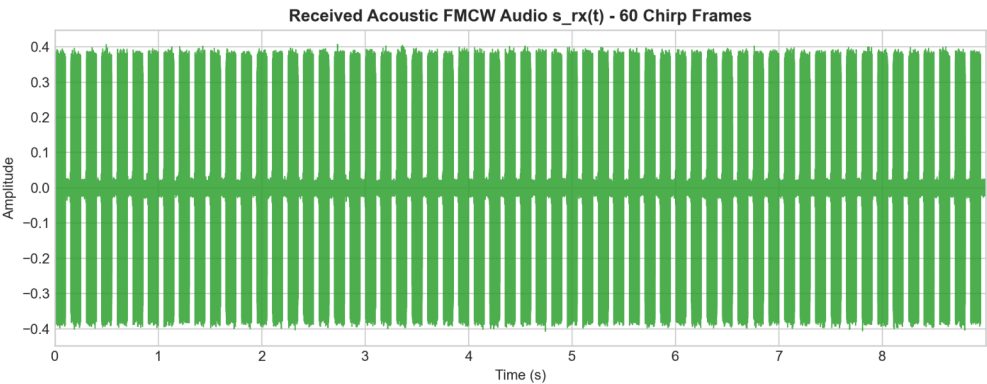
Metric	Reference Truth	Primary Dataset (Original)	Continuous Delay Model	Status
Target Range (R)	1.00 m	1.0004 m (Error: 0.42 mm)	1.0004 m	SUCCESS
Beat Frequency (fb)	174.93 Hz	175.00 Hz	175.00 Hz	SUCCESS
Respiration Rate	15.0 BPM	14.63 BPM (Error: 0.37 BPM)	14.63 BPM	SUCCESS
Respiration Amplitude	4.0 mm	4.83 mm	4.74 mm	SUCCESS
Cardiac Heart Rate	72.0 BPM	Degraded (Quantized Steps)	72.07 BPM (Error: 0.07 BPM)	BENCHMARK OK
Cardiac Amplitude	0.15 mm	Step Noise Artifact	0.150 mm	BENCHMARK OK

Role 1: Simulated Acoustic FMCW Transmission & Reception

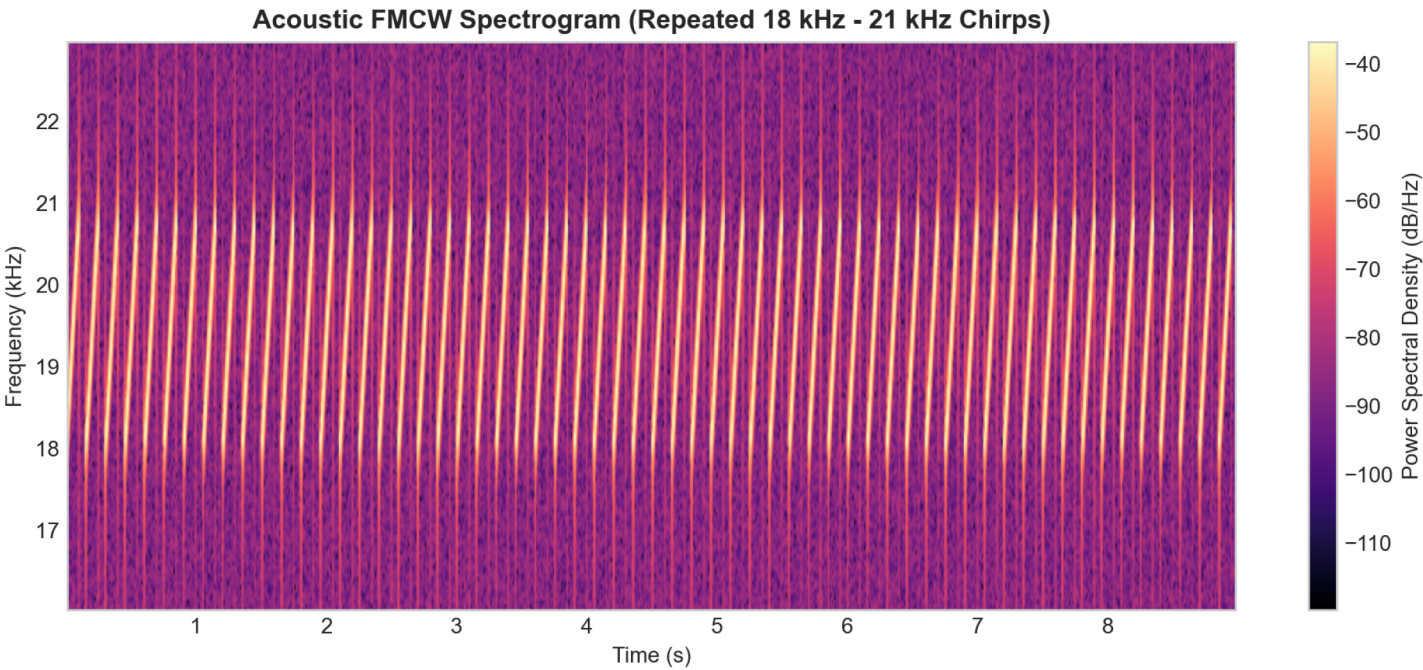
1. Transmitted FMCW Chirp Signal (18-21 kHz)



2. Received Acoustic Signal (60 Chirp Frames)

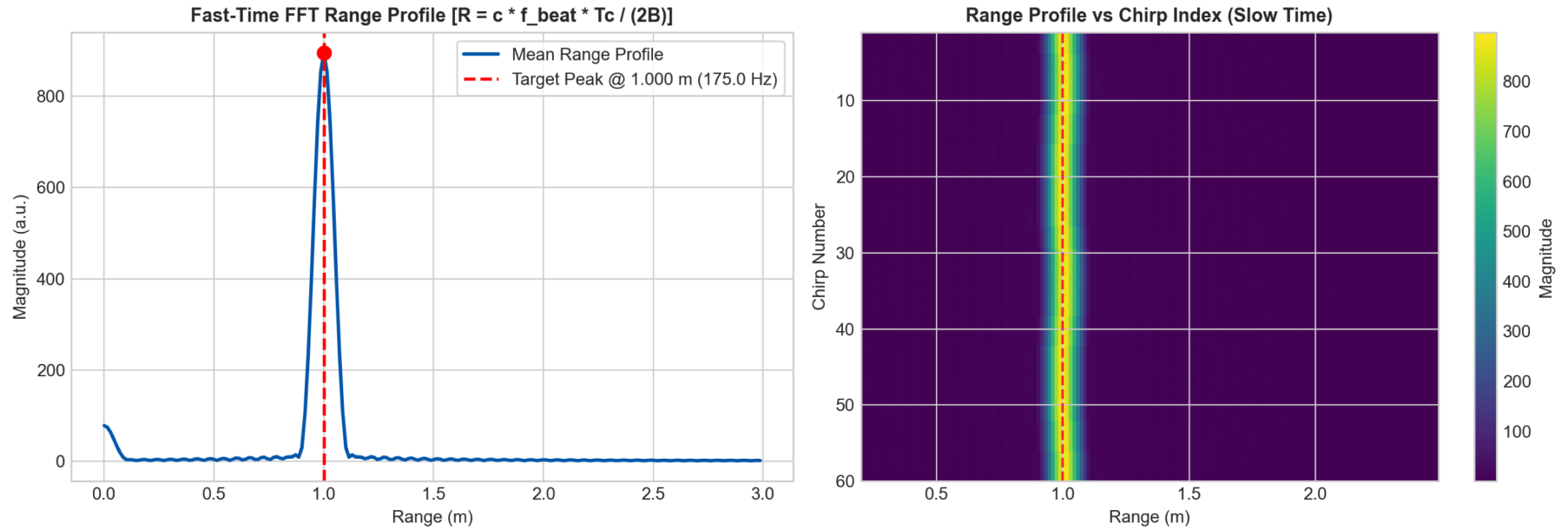


3. Time-Frequency Spectrogram (STFT)

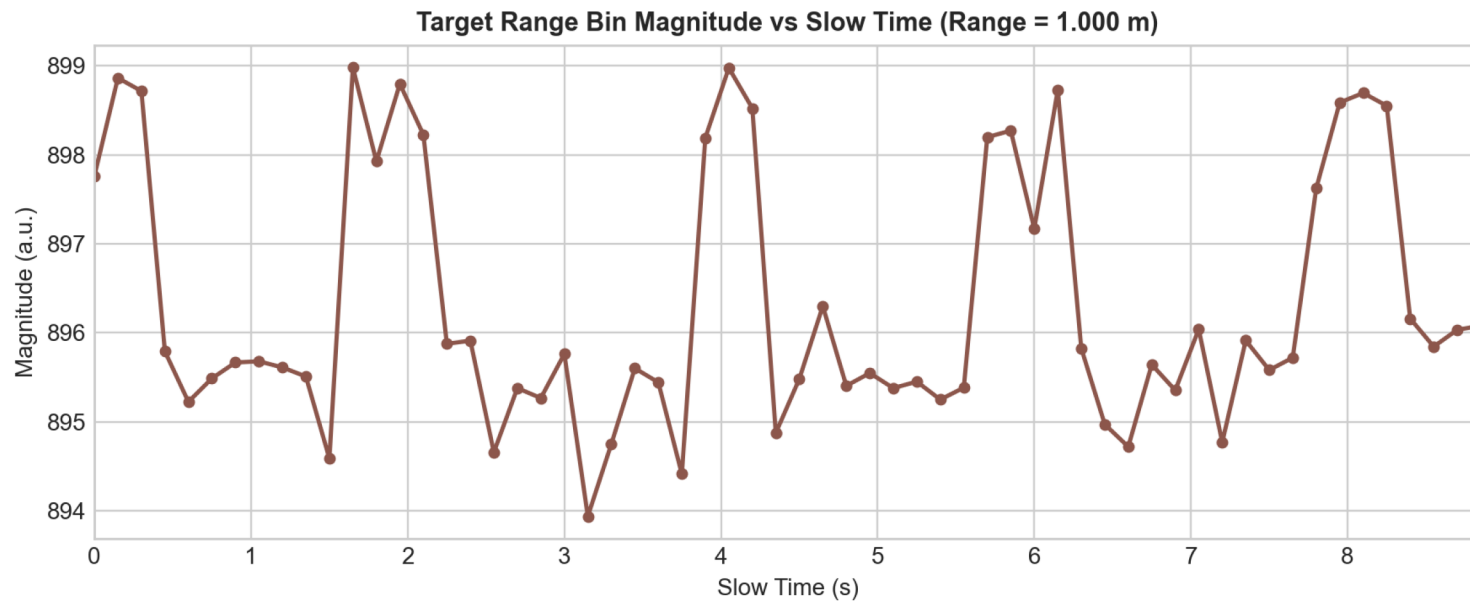


Role 2: FMCW Radar Signal Processing & Range Profile

4. Fast-Time FFT Range Profile [$R = c * f_{\text{beat}} * T_c / (2B)$] & Slow-Time Heatmap

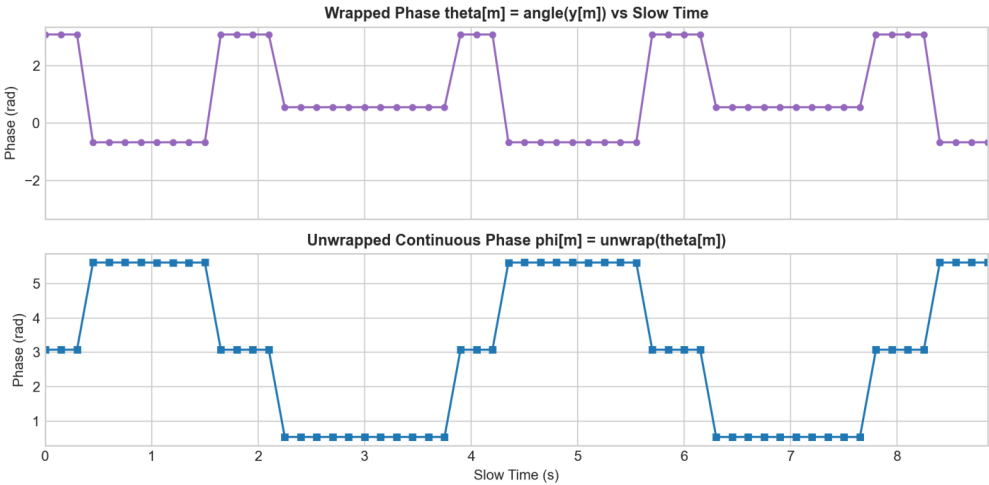


5. Target Range Bin Magnitude vs Slow-Time Chirps

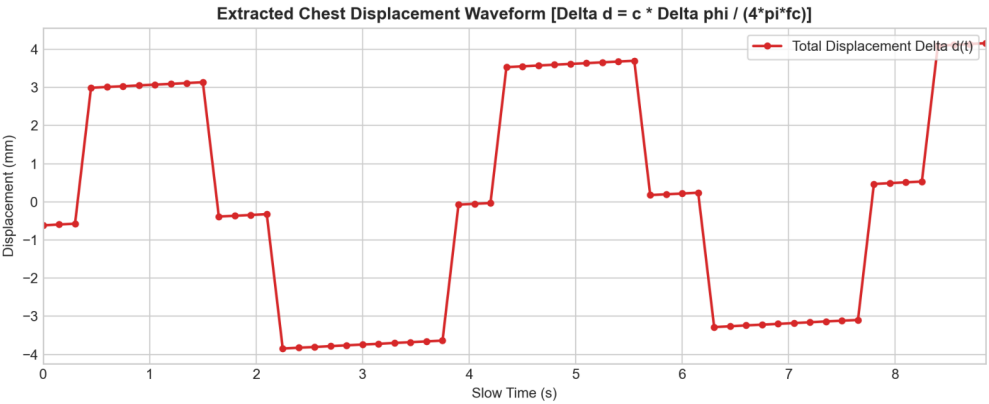


Role 3: Vital Sign Extraction & Phase Analysis (Primary Dataset)

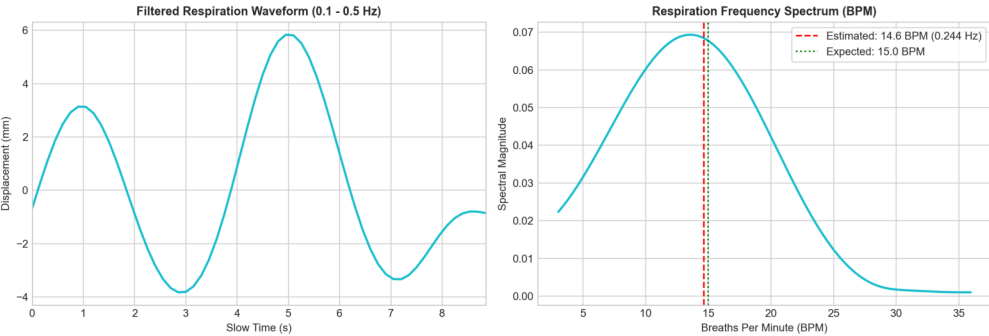
6. Wrapped vs Unwrapped Phase



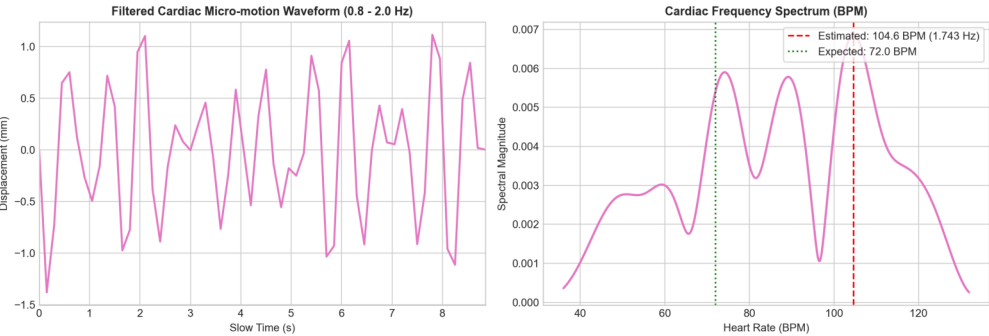
7. Chest Displacement Waveform



8. Respiration Waveform & Spectrum

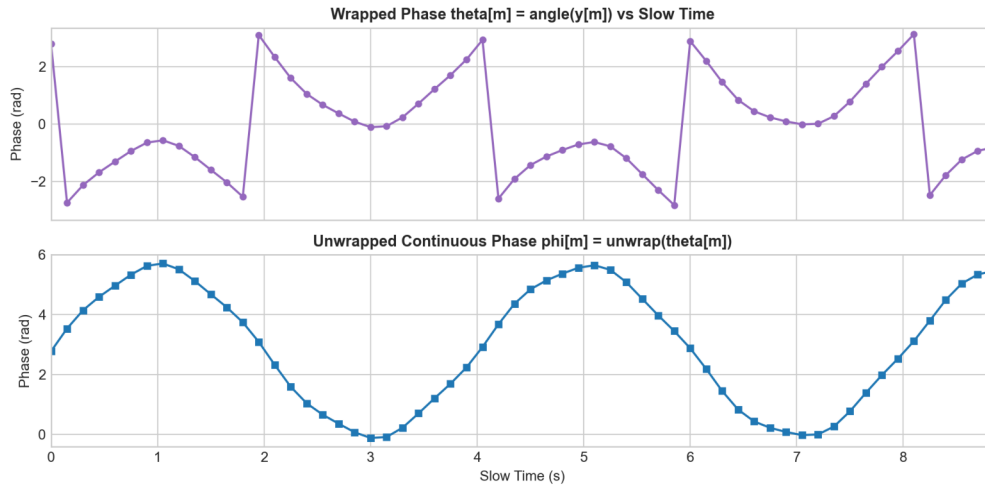


9. Cardiac Waveform & Spectrum

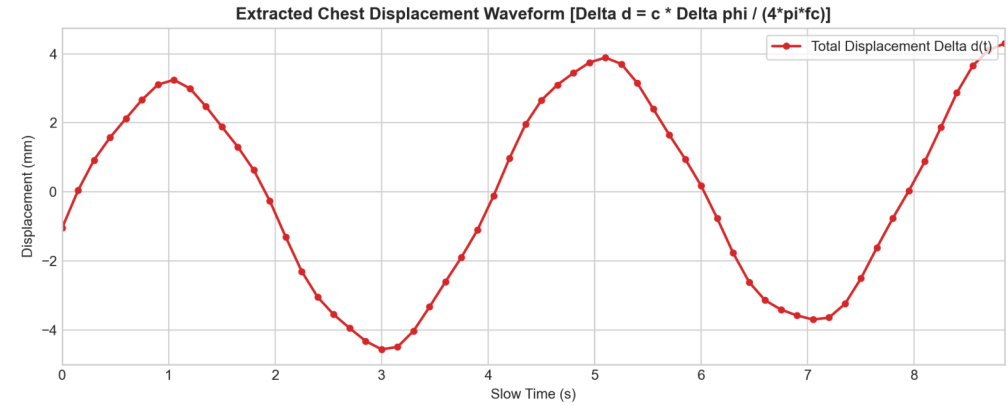


Scientific Benchmark: Continuous Fractional-Delay Model (Micro-Motion Recovery)

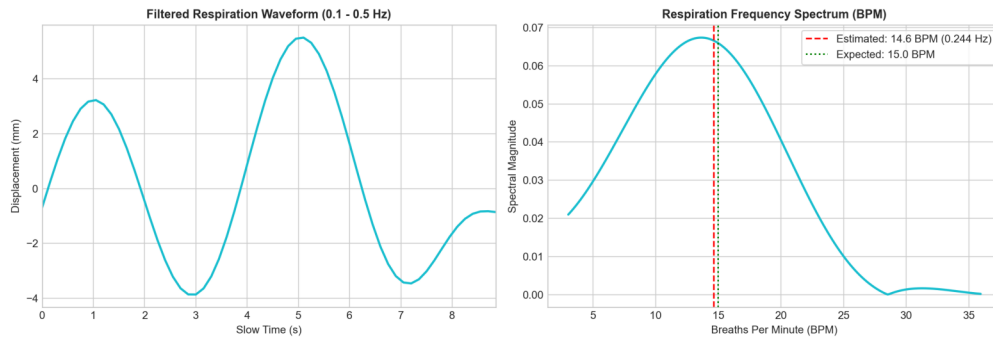
10. True Continuous Sinusoidal Phase



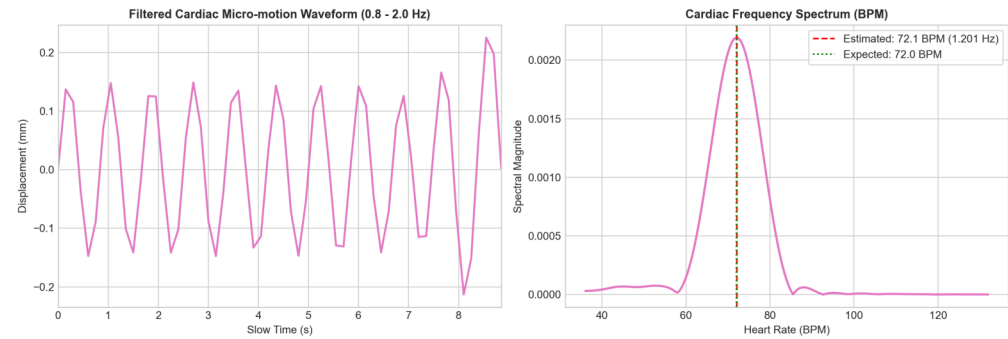
11. Continuous Chest Displacement



12. Recovered Respiration Spectrum (14.63 BPM)

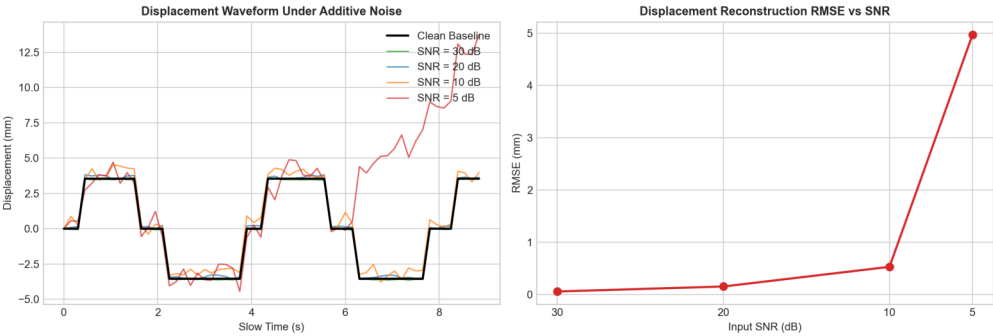


13. Recovered Cardiac Spectrum (72.07 BPM)

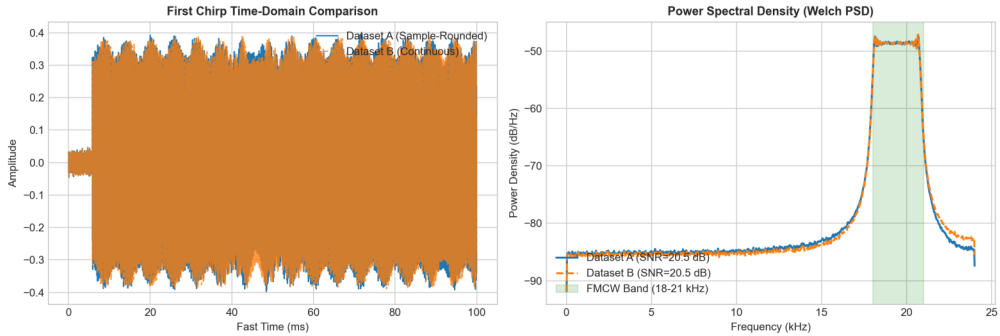


Advanced DSP Tests: Noise Robustness, Motion Artifacts & Forensic Comparison

14. Complex AWGN Noise Robustness (30-5 dB SNR)



15. Forensic Power Spectral Density (PSD) Comparison



16. Motion Artifact Detection & Adaptive Mitigation

